Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam B · Form 1: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. Release 3 kg and 1 kg masses from rest on an ideal pulley. Use g=10 m/s². Where is tension relative to the weights?

(A) T=10 N

(B) T>40 N

(C) T=30 N

(D) 10 N<T<30 N

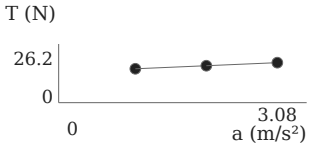
2. The left mass on an ideal pulley is 2 kg; vary the right mass. Left upward is positive; use g=10 m/s². a (m/s²) -> T (N): (1.08, 22.16); (2.08, 24.16); (3.08, 26.16). Predict the reading at horizontal coordinate 2.58.

(A) 15.1

(B) 25.16

(C) 35.22

(D) 45.29



3. An Atwood machine starts from rest. The heavier mass descends. A spring scale inserted into the massless rope reads T. Which inequality is correct?

(A) T is below both weights

(B) T exceeds both weights

(C) Light weight < T < heavy weight

(D) T equals the heavy weight

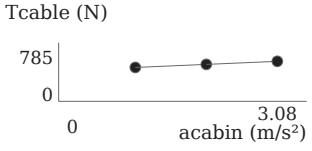
4. Release a 60 kg cabin and adjustable counterweight over an ideal pulley without a motor. Cabin up is positive; g=10 m/s². acabin (m/s²) -> Tcable (N): (1.08, 664.8); (2.08, 724.8); (3.08, 784.8). What is the slope between the endpoint measurements?

(A) 60

(B) 84

(C) 36

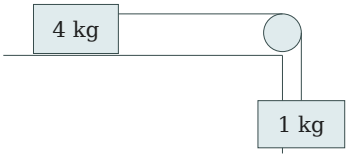
(D) 108



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Release a frictionless 4 kg cart connected to a 1 kg hanger on an ideal pulley. Use g=10 m/s².

a. Label the horizontal force on the cart. **(1 pt)**



b. Determine the common acceleration magnitude and the tension. **(3 pt)**

6. Transfer mass between a frictionless cart and its hanger on an ideal pulley. Total mass is 5 kg; g=10 m/s². mh (kg) -> a (m/s²): (0.1296, 0.2592); (0.2496, 0.4992); (0.3696, 0.7392).

a. Build the model for track and hanger from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A heavier mass H and a lighter mass L hang over an ideal pulley. Derive the tension without approximating it as either weight. **(3 pt)**